

CODEALPHA DATA SCIENCE INTERNSHIP

TASK 3 - DATA VISUALIZATION

Project Report & Python Implementation

Intern Name	_____
College / Institute	_____
Branch	AI & Data Science (AIDS)
Task	Task 3 - Data Visualization
Date	10 September 2026

Task Objective: Transform raw data into meaningful visual formats such as charts and graphs, use visualization libraries, reveal patterns clearly, and communicate data-driven insights.

1. Introduction

Data visualization is the process of representing data graphically so that trends, patterns, comparisons, and unusual observations can be understood easily. In this task, a book dataset collected through web scraping is used for visualization.

Tools Used

- Python
- Pandas for data handling
- Matplotlib for creating charts
- Jupyter Notebook / Google Colab

2. Dataset Description

The dataset contains book information such as title, price, availability, rating, and URL. Before visualization, the price is converted from text into a numerical value.

Column	Type	Purpose
title	Text	Book name
price	Text	Original displayed price
price_num	Numeric	Clean price used for charts
availability	Text	Availability status
rating	Numeric	Book rating from 1 to 5

3. Data Preparation

The following steps prepare the dataset before plotting:

- Load the CSV file using Pandas.
- Remove duplicate records.
- Convert price values into numeric format.
- Check missing values.
- Use the cleaned columns for visualization.

Python Code

```
import pandas as pd

df = pd.read_csv("books_scraped.csv")

# Remove duplicate rows
df = df.drop_duplicates()

# Convert price to numeric
df["price_num"] = (
    df["price"]
    .str.replace("£", "", regex=False)
    .astype(float)
)

# Check missing values
print(df.isnull().sum())

print(df.head())
```

4. Visualization 1 - Price Distribution

A histogram is used to understand how book prices are distributed. It groups prices into ranges and shows how many books fall within each range.

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(7, 4))
plt.hist(df["price_num"], bins=15)
plt.title("Distribution of Book Prices")
plt.xlabel("Price")
plt.ylabel("Number of Books")
plt.tight_layout()
plt.show()
```

Insight: The histogram helps identify the most common price range and whether the prices are concentrated or widely spread.

5. Visualization 2 - Rating Distribution

A bar chart is used to compare the number of books receiving each rating from 1 to 5 stars.

```
rating_counts = df["rating"].value_counts().sort_index()
```

```
plt.figure(figsize=(7, 4))
plt.bar(rating_counts.index.astype(str), rating_counts.values)
plt.title("Book Rating Distribution")
plt.xlabel("Rating (Stars)")
plt.ylabel("Number of Books")
plt.tight_layout()
plt.show()
```

Insight: The chart makes it easy to identify the most frequent rating and compare rating categories.

6. Visualization 3 - Price Box Plot

A box plot summarizes the spread of book prices and helps identify unusually high or low values (possible outliers).

```
plt.figure(figsize=(7, 3.5))
plt.boxplot(df["price_num"].dropna(), vert=False)
plt.title("Book Price Spread")
plt.xlabel("Price")
plt.tight_layout()
plt.show()
```

Insight: The box plot shows the median, overall spread, and possible outliers in the book-price data.

7. Visualization 4 - Top 10 Most Expensive Books

A horizontal bar chart is used to compare the ten books with the highest prices.

```
top10 = df.nlargest(10, "price_num").sort_values("price_num")
```

```
plt.figure(figsize=(8, 5))
plt.barh(top10["title"], top10["price_num"])
plt.title("Top 10 Most Expensive Books")
plt.xlabel("Price")
plt.tight_layout()
plt.show()
```

Insight: This chart provides a direct comparison of the highest-priced books and makes ranking easy to understand.

8. Visualization Summary

Visualization	Main Purpose	Insight
Histogram	Price distribution	Shows common price ranges
Bar chart	Rating comparison	Shows frequency of each rating
Box plot	Price spread	Shows median, spread and outliers
Horizontal bar chart	Top 10 prices	Ranks expensive books

Good visualizations should have a clear title, meaningful axis labels, readable text, and a chart type appropriate for the question being answered.

9. Conclusion

Task 3 demonstrates how raw data can be transformed into useful visual information. Histograms show distributions, bar charts support comparisons, box plots reveal spread and outliers, and ranking charts communicate top values. These visualizations make data easier to interpret and support data-driven decision-making.

Submission Checklist

- ☐ Run all code in Jupyter Notebook or Google Colab.
- ☐ Verify that the charts are generated successfully.
- ☐ Take screenshots of the code and charts if required.
- ☐ Submit this PDF together with the required project files.